SYSDESIGN REFERENCE

Multi-Region Architecture

A senior-level reference with tradeoffs, failure modes, and interview follow-ups for multi-region architecture.

Place computation and data across regions while explicitly choosing consistency, routing, and disaster behavior.

VERSION 1.0 REVIEW
REVIEWED THROUGH 2026-07-18
STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / MULTI-REGION

SYSDESIGN

Multi-Region Architecture

CORE IDEA

Place computation and data across regions while explicitly choosing consistency, routing, and disaster behavior.

Reference architecture decision chain

01	02	03	04
Active-Active Vs Active-Passive	Data Ownership	Replication	Failover

Decisions to defend

+ Active-active vs active-passive
+ data ownership
+ replication
+ failover
+ and residency.

Failure modes to test

+ Split brain
+ stale DNS
+ dependency locality
+ and untested failback.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Multi-Region Architecture

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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